21				5	M1 $\overrightarrow{BA} = \mathbf{a} - 3\mathbf{b}$ or $\overrightarrow{YA} = \frac{1}{2}(\mathbf{a} - 3\mathbf{b})$
					M1 $\overrightarrow{XY} = \mathbf{b} + \frac{1}{2}(\mathbf{a} - 3\mathbf{b})$ or $\overrightarrow{YZ} = \mathbf{a} + \frac{1}{2}(\mathbf{a} - 3\mathbf{b})$ or $\overrightarrow{XZ} = 2\mathbf{a} - 2\mathbf{b}$ M1 two of $\overrightarrow{XY} = \mathbf{b} + \frac{1}{2}(\mathbf{a} - 3\mathbf{b})$ or $\overrightarrow{YZ} = \mathbf{a} + \frac{1}{2}(\mathbf{a} - 3\mathbf{b})$ or $\overrightarrow{XZ} = 2\mathbf{a} - 2\mathbf{b}$ M1 simplification to enable comparison of 2 of $\overrightarrow{XY}$, $\overrightarrow{XZ}$ and $\overrightarrow{YZ}$ e.g. 2 of $\frac{1}{2}(\mathbf{a} - \mathbf{b})$, $\frac{3}{2}(\mathbf{a} - \mathbf{b})$ **and** $2(\mathbf{a} - \mathbf{b})$ C1 completes proof with reference to 2 of $\overrightarrow{XY}$, $\overrightarrow{XZ}$ and $\overrightarrow{YZ}$ parallel and a common point

Documented scheme clarification

1MA1_MOCK_SET_01_2H_Q21

Original source

scheme page 13, SHA256 6cc909fae33e027799cde0a5288d0ba8ae2bff761ed5d978aed4248c309b584f

Original wording

The later repeated XY method line prints -b + 1/2(a - 3b).

Correction / interpretation

Use XY = XB + BY = b + 1/2(a - 3b), which simplifies to 1/2(a - b). The later repeated method line prints -b + 1/2(a - 3b); that minus sign is a misprint. YZ = 3/2(a - b) and XZ = 2(a - b), so the vectors are parallel and share points.

Reason

XB = b and BY = 1/2 BA = 1/2(a - 3b). The first method line and final simplified vectors are correct; the later repeated XY line must start with positive b.

SoI authority

255ab3b5e61075efcd15bf38f84a9b1c3db32548995214bca495d2a2935b5569